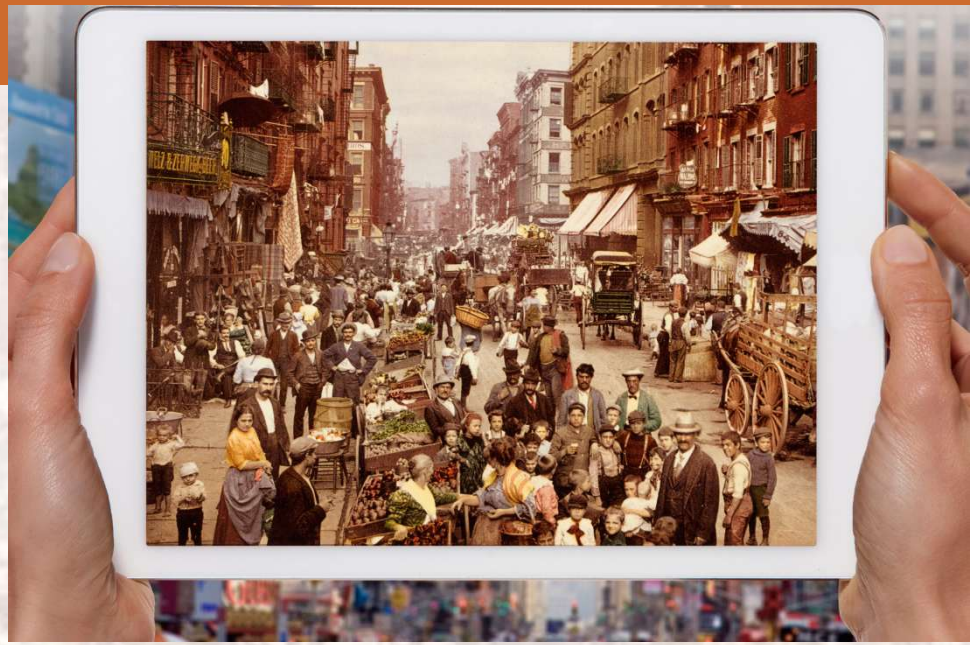
# 經濟學



CHAPTER

1

# Ten Principles of Economics

National Central University

# Look for the answers to these questions:

- What kinds of questions does economics address?
- What are the principles of how people make decisions? (e.g., *Marginal thinking*)
- What are the principles of how people interact? (e.g., *Eye for an Eye*)
- What are the principles of how the economy as a whole works?



# Ten Principles of Economics

- Resources are scarce 資源有限,慾望無窮
- Scarcity: the limited nature of society's resources
  - Society has limited resources
    - Cannot produce all the goods and services people wish to have
- Economics
  - The study of how society manages its scarce resources (供給=需求; 均衡)



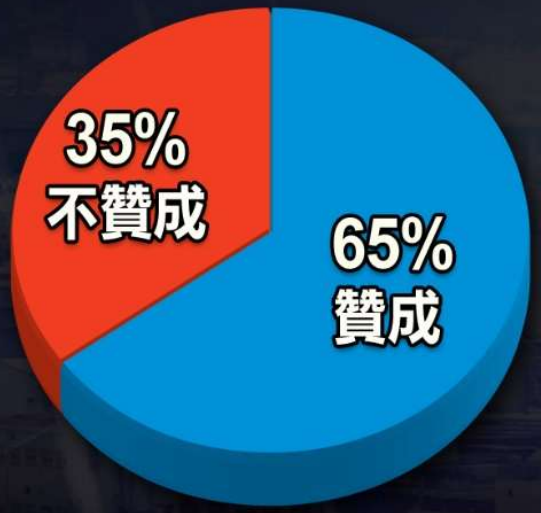
# Ten Principles of Economics

- Economists study:
  - How people decide what to buy, how much to work, save, and spend
    預算有限
    消費vs休閒
    現在享樂vs未來享樂
  - How firms decide how much to produce, how many workers to hire
    生產力
  - How society decides how to divide its resources between national defense, consumer goods, protecting the environment, and other needs



# 台化遷廠爭議

縣府宣布台化停工，贊成?




停工致使勞工失業，適當?




註：調查時間：9/22~9/23有效問卷：1108份

資料來源：彰化縣政府計畫處



# 台化遷廠爭議

## 台化彰化廠爭議 土地開發才是主因？

| 台塑集團總管理處總經理 王瑞瑜 | 彰化縣長 魏明谷 |
| --- | --- |
| ■ 台化曾承諾前任縣府，彰化廠土地47%回饋縣府辦都更、53%保留自用生產，新縣府上台後就要100%（都更） | ■ 鍋爐操作許可沒有通過，是因為沒有符合環保審查承諾，跟土地開發無關 |
| ■ 政府只是要我們的土地，就要把我們趕出去 | ■ 原址都更後，須100%作商業區使用。執意保留工廠生產，不符合縣民期待。台化沒有理由留在彰化市區 |

有話好說整理



# 台化遷廠爭議





# 張善政交通政見出爐公車、YouBike享30公里免費里程




How would you assess this policy?



# 張善政交通政見出爐公車、YouBike享30公里免費里程

2021/08/03 00:27

〔記者魏瑾筠／桃園報導〕桃園市自2016年起推動YouBike（微笑單車），並祭出前30分鐘內免費騎乘優惠，使用次數年年攀升，但審計部桃園市審計處發現，市府補貼給YouBike的費用從2016年逾6200萬元，到去年已增加至1.1億元，且逾9成民眾租車不到30分鐘，還有使用者單日多次借還以規避付費，桃園市審計處認為，應讓YouBike回歸使用者付費，減輕桃市府財政負擔。桃園市政府交通局回應，考量捷運綠線施工中，且YouBike場站正陸續建置，為增加民眾對YouBike的黏著度，暫不考慮收費。

2023/03/09 05:30

## 目前407站萬餘輛車 9成騎免錢

桃園市境現有YouBike四○七站、一萬多輛自行車，建置地點位於大眾運輸場站、學校、商圈等地，交通局統計桃園、中壢都會區的YouBike每日平均周轉率約十五次，其餘地區近三次，目前費率是前四小時每半小時十元，前卅分鐘不限次數或卡種免費，由市府補貼，以往每年租金補貼約一·二到一·四億元，以十元差額評估，約要追加一二○○萬元預算。



# 解貧富差距 侯友宜喊基本薪資3.3萬





# How People Make Decisions

Principle 1: People face trade-offs 比較的哲學

Principle 2: The cost of something is what you give up to get it 機會成本

Principle 3: Rational people think at the margin 邊際價值 = 邊際成本

Principle 4: People respond to incentives



# Principle 1: People Face Trade-offs

- To get something that we like, we have to give up something else that we also like
  - Going to a party the night before an exam
    - Less time for studying
  - Having more money to buy stuff 慾望的奴隸
    - Working longer hours, less time for leisure
  - Protecting the environment 資源有限
    - Resources could be used to develop urban transportation services



# Principle 1: People Face Trade-offs

- Society faces trade-offs:
  - The more it spends on national defense (guns) to protect its shores
    - The less it can spend on consumer goods (butter) to raise the standard of living at home
  - Pollution regulations: cleaner environment and improved health
    - But at the cost of reducing the incomes of the firms' owners, workers, and customers

保護環境造成商品成本上揚,員工福利減少



# Principle 1: People Face Trade-offs

- Efficiency: society gets the most from its scarce resources 資優班
- Equality: prosperity is distributed uniformly among society's members 均富
- Tradeoff:
  - To achieve greater equality, could redistribute income from wealthy to poor
  - But this reduces incentive to work and produce, shrinks the size of economic "pie"

高稅額會降低工作意願



# Principle 2: The Cost of Something Is What You Give Up to Get It

- Making decisions:
  - Compare costs with benefits of alternatives
  - Need to include opportunity costs
- Opportunity cost
  - Whatever must be given up to obtain some item



# Principle 2: The Cost of Something Is What You Give Up to Get It

- The opportunity cost of:
  - Going to college for a year
    - Tuition, books, and fees
    - PLUS foregone wages
  - Going to the movies
    - The price of the movie ticket
    - PLUS the value of the time you spend in the theater



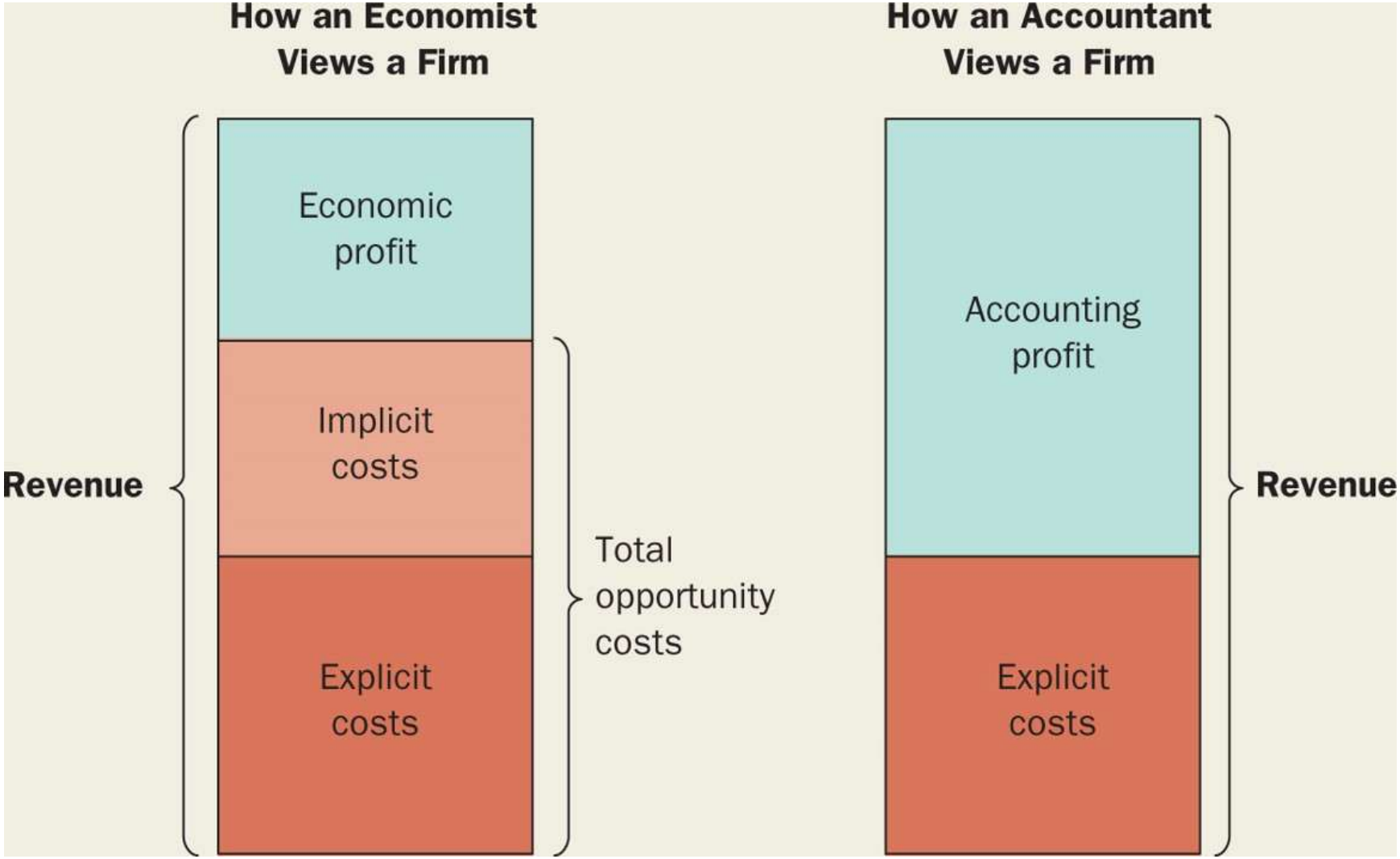
# Question

- You win $100 in a basketball pool.
- You have a choice between spending the money now and putting it away for a year in a bank account that pays 5 percent interest .
- What is the opportunity cost of spending the $100 now?





# 機會成本與會計成本差異





# Principle 3: Rational People Think at the Margin

- Rational people
  - Systematically and purposefully do the best they can to achieve their objectives
  - Given the available opportunities

    在可用的機會裡做出選擇

  - Make decisions by evaluating costs and benefits of marginal changes
    - Small incremental adjustments to a plan of action

      e.g., スシロー



# Principle 3: Rational People Think at the Margin

- Examples: 類似的例子: 飛機
  - Cell phone users with unlimited minutes (the minutes are free at the margin)
    - Are often prone to making long/frivolous calls
    - Marginal benefit of the call > 0 沒事也會打電話

      為什麼水是必需品,但價格比鑽石便宜?
  - A manager considers whether to increase output
    - Compares the cost of the needed labor and materials to the extra revenue



# Question

- The company that you manage has invested $5 million in developing a new product, but the development is no quite finished.
- At a recent meeting, your salespeople report that the introduction of competing products has reduced the expected sales of your new product to $3 million.
- If it would cost $1 million to finish development and make the product, should you go ahead and do so?
- What is the most that you should pay to complete development



# Principle 4: People Respond to Incentives

- Incentive
  - Something that induces a person to act
- Examples:
  - When gas prices rise, consumers buy more hybrid cars and fewer gas guzzling SUVs
  - When cigarette taxes increase, teen smoking falls

為什麼根據報導,安全帶推廣之後,駕駛死亡率仍然沒有顯著下降呢?



# 車款介紹




*Ford Mustang*

# Active Learning 1 — Applying the principles

You are selling your 2007 Mustang. You have already spent $1,000 on repairs. At the last minute, the transmission dies. You can pay $900 to have it repaired, or sell the car "as is."

In each of the following scenarios, should you have the transmission repaired? Explain.

A. Blue book value (what you could get for the car) is $7,500 if transmission works, $6,200 if it doesn't.

B. Blue book value is $6,300 if transmission works, $5,500 if it doesn't.

# Active Learning 1 — Answers

Cost of fixing the transmission = $900

A. Blue book value is $7,500 if transmission works, $6,200 if it doesn't

- Benefit of fixing transmission = $1,300 (= 7500 – 6200)
- Get the transmission fixed

B. Blue book value is $6,300 if transmission works, $5,500 if it doesn't

- Benefit of fixing the transmission = $800 (= 6300 – 5500)
- Do not pay $900 to fix it



# Question

- A 1996 bill reforming the federal government's antipoverty programs limited many welfare recipients to only two years of benefits
  - A. How does this change affect the incentives for working?
  - B. How might this change represent a trade-off between equality and efficiency?



# How People Interact

Principle 5: Trade can make everyone better off

Principle 6: Markets are usually a good way to organize economic activity

Principle 7: Governments can sometimes improve market outcomes



# Principle 5: Trade Can Make Everyone Better Off

- People benefit from trade:
  - People can buy a greater variety of goods and services at lower cost
- Countries benefit from trade and specialization 專業分工
  - Get a better price abroad for goods they produce 賣高
  - Buy other goods more cheaply from abroad than could be produced at home 買低



# 雞蛋不要放進同一個籃子

112年3至7月進口雞蛋專案流向表

| 進口國家 | 進口數量(萬顆) | 合計(萬顆) | 流向情況 | | 數量(萬顆) | 比例 |
|---|---|---|---|---|---|---|
| 泰國 | 2,185 | 14,531 | 洗選 | | 2,491 | 17.14% |
| 菲律賓 | 179 | | | | | |
| 美國 | 373 | | 加工 | | 3,428 | 23.59% |
| 土耳其 | 2,246 | | 庫存 | 殼蛋 | 625 | 4.30% |
| 澳洲 | 591 | | | 加工 | 2,584 | 17.78% |
| 巴西 | 8,891 (61%) | | 耗損與到期 | 已銷毀 | 232 | 1.60% |
| 馬來西亞 | 65 | | | 待銷毀 | 5,170 | 35.58% |

- 耗損與到期共5,402萬顆，約全國2天需求量
- 到期蛋品以堆肥再利用，不會流入國內市場，為消費者把關，也避免影響國內蛋農

11

United Daily News(Sep 4, 2023)
藍營批巴西5月就公告為禽流感地區中央還讓超思進口蛋



# Principle 6: Markets Are Usually a Good Way to Organize Economic Activity

- Market
  - A group of buyers and sellers (need not be in a single location)
- "Organize economic activity" means determining
  - What goods and services to produce
  - How much of each to produce
  - Who produced and consumed these

家計單位的決策 vs. 公司行號的決策



# Principle 6: Markets Are Usually a Good Way to Organize Economic Activity

- A market economy allocates resources
  - Decentralized decisions of many firms and households – as they interact in markets
- Famous insight by Adam Smith in The Wealth of Nations (1776):

價格機制

  - Each of these households and firms acts as if "led by an invisible hand" to promote general economic well-being

你的本意並非如此,但只能接受市場的價格



# Principle 6: Markets Are Usually a Good Way to Organize Economic Activity

- Prices:
  - Determined: interaction of buyers and sellers
  - Reflect the good's value to buyers
  - Reflect the cost of producing the good
- Invisible hand:
  - Prices guide self-interested households and firms to make decisions that maximize society's economic well-being



# Principle 7: Governments Can Sometimes Improve Market Outcomes

- Government - enforce property rights
  - Enforce rules and maintain institutions that are key to a market economy
    - People are less inclined to work, produce, invest, or purchase if large risk of their property being stolen

如果沒有法律與執行機構
1.沒人敢出國去旅遊(侵占)
2.餐廳無法經營(賴帳)
3.沒人拍電影(盜版)



# Principle 7: Governments Can Sometimes Improve Market Outcomes

- Government - promote efficiency
  - Avoid market failures: market left on its own fails to allocate resources efficiently
  - Externality – source of market failure 市場失靈
    - Production or consumption of a good affects bystanders (e.g. pollution) 躺著也中槍
  - Market power – source of market failure
    - A single buyer or seller has substantial influence on market price (e.g. monopoly) 價格被操控



# Principle 7: Governments Can Sometimes Improve Market Outcomes

- Government - promote equality
  - Avoid disparities in economic wellbeing
  - Use tax or welfare policies to change how the economic “pie” is divided

Economic wellbeing: 經濟福祉/福利
Tax: 財富重分配
Welfare policies: 食物券,三倍券

# Active Learning 2 — Discussion Question

In each of the following situations, what is the government's role?

Does the government's intervention improve the outcome?

a. Public schools for K-12
b. Workplace safety regulations
c. Public highways (Public transportation)
d. Patent laws, which allow drug companies to charge high prices for life-saving drugs



# Question

- Explain whether each of the following government activities is motivated by a concern about equality or a concern about efficiency.
- In the case of efficiency, discuss the type of market failure involved.
  - A. regulating cable TV prices
  - B. providing some poor people with vouchers that can be used to buy food
  - C. prohibiting smoking in public places
  - D. breaking up Standard Oil (which once owned 90% of all oil refineries) into several smaller companies
  - E. imposing higher personal income tax rates on people with higher incomes
  - F. Instituting laws against driving while intoxicated



# How the economy as a whole works

Principle 8: A country's standard of living depends on its ability to produce goods and services

Principle 9: Prices rise when the government prints too much money

Principle 10: Society faces a **short-run** trade-off between inflation and unemployment



# Principle 8: Country's Standard of Living Depends on Its Ability to Produce Goods and Services

- Huge variation in living standards
  - Across countries and over time
  - Average income in rich countries
    - Is more than ten times average income in poor countries
  - The U.S. standard of living today
    - Is about eight times larger than 100 years ago



# Principle 8: Country's Standard of Living Depends on Its Ability to Produce Goods and Services

- Productivity: most important determinant of living standards
  - Quantity of goods and services produced from each unit of labor input
  - Depends on the equipment, skills, and technology available to workers
    - Other factors (e.g., labor unions, competition from abroad) have far less impact on living standards



# Principle 9: Prices Rise When the Government Prints Too Much Money

- Inflation
  - An increase in the overall level of prices in the economy
- In the long run
  - Inflation is almost always caused by excessive growth in the quantity of money, which causes the value of money to fall
  - The faster the government creates money, the greater the inflation rate



# Principle 10: Society Faces a Short-run Trade-off between Inflation and Unemployment

- Short-run trade-off between unemployment and inflation
  - Over a period of a year or two, many economic policies push inflation and unemployment in opposite directions
  - Other factors can make this tradeoff more or less favorable, but the tradeoff is always present

印鈔票→消費增加→物價上漲→廠商增產→聘用員工 (通貨膨脹增加,失業率下跌)

長期來看,失業率與印鈔票無關聯



# Question

- During the Revolutionary War, the American colonies could not raise enough tax revenue to fully fund the war effort.
- To make up the difference, the colonies decided to print more money.
- Printing money to cover expenditures is sometimes referred to as an “inflation tax (通貨膨脹稅)”.
- Who do you think is being “taxed” when more money is printed? Why?



# You're in School!
# Did You Make the Right Decision?





# Eye on the Benefit and Cost of School (1 of 2)

- **Did You Make the Right Decision?**
  - Does school provide a big enough benefit to justify its cost?
  - The **benefits** of being in school include the present enjoyment of student life and a higher future income.
  - The **costs** of being in school for a full-time student include tuition, books, other study costs, and forgone earnings.
- Is school always best?
  - The costs are incurred now, but the benefits accrue over a working lifetime.
- For most people, the net benefit is big!



# Eye on the Benefit and Cost of School (2 of 2)

- **Did You Make the Right Decision?**
- Is school always best?
  - Bill Gates quit Harvard
  - Mark Zuckerberg quit Harvard
  - Clayton Kershaw turned down a scholarship at Texas A & M! (德州農工大學)
- All three expected the benefit from school to be less than the opportunity cost of being in school.






# Summary

- Fundamental lessons about individual decision making:
  - People face trade-offs among alternative goals
  - The cost of any action is measured in terms of forgone opportunities
  - Rational people make decisions by comparing marginal costs and marginal benefits
  - People change their behavior in response to the incentives they face

# Summary

- Fundamental lessons about interactions among people:
  - Trade and interdependence can be mutually beneficial
  - Markets are usually a good way of coordinating economic activity among people
  - The government can potentially improve market outcomes by remedying a market failure or by promoting greater economic equality

# Summary

- Fundamental lessons about the economy as a whole:
  - Productivity is the ultimate source of living standards
  - Growth in the quantity of money is the ultimate source of inflation
  - Society faces a short-run trade-off between inflation and unemployment